

Proteomics, Metabolomics and Multi-Omics



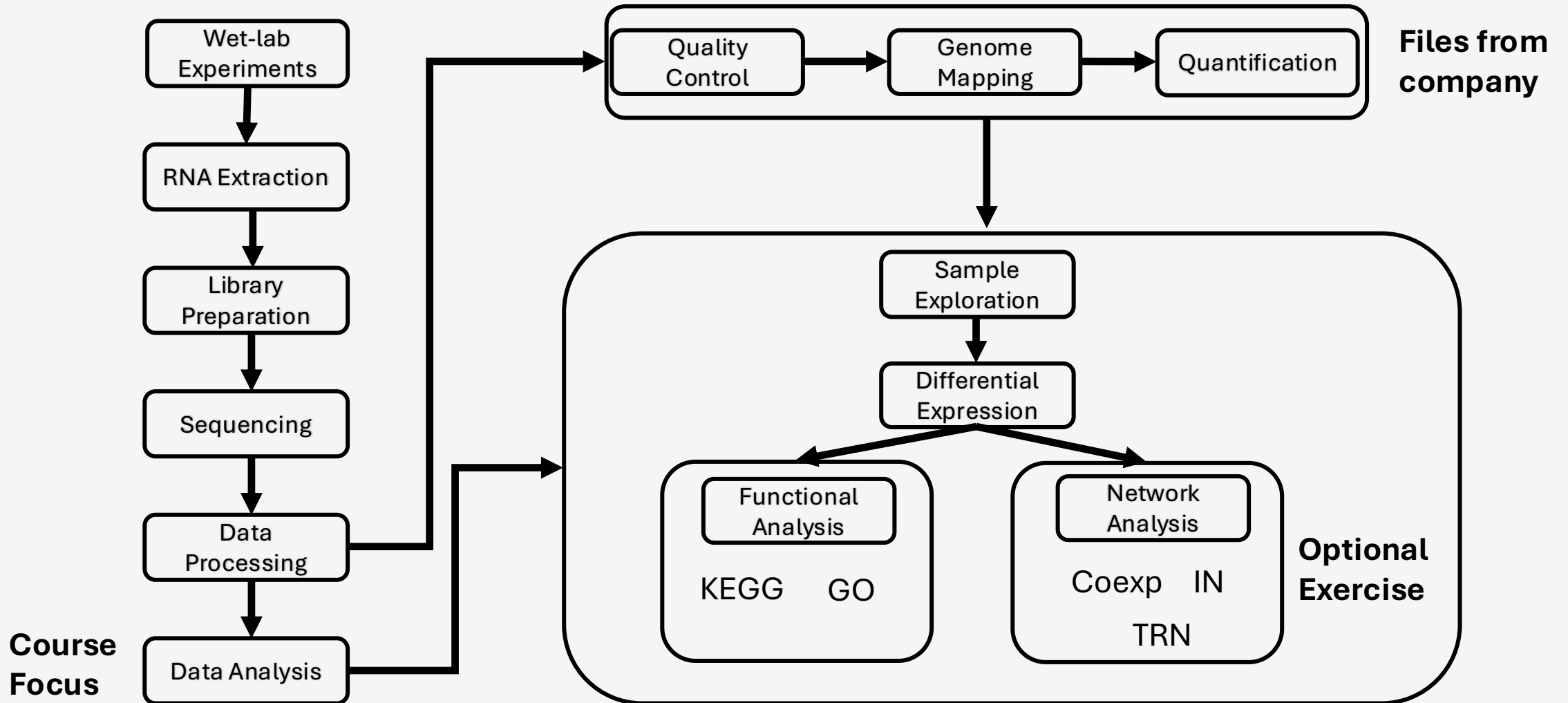
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Recap



What becomes visible after RNA



potential → machinery → biochemical state

Today is not “RNA-seq again”.
We focus on what **proteins and metabolites** reveal that RNA cannot.

What is new today?

PROTEOMICS

Protein abundance, turnover, complexes, isoforms, post-translational modifications, dynamic range.

METABOLOMICS

Small molecules, pathway output, diet/drug/environment signals, fast dynamics, annotation uncertainty.

INTEGRATION

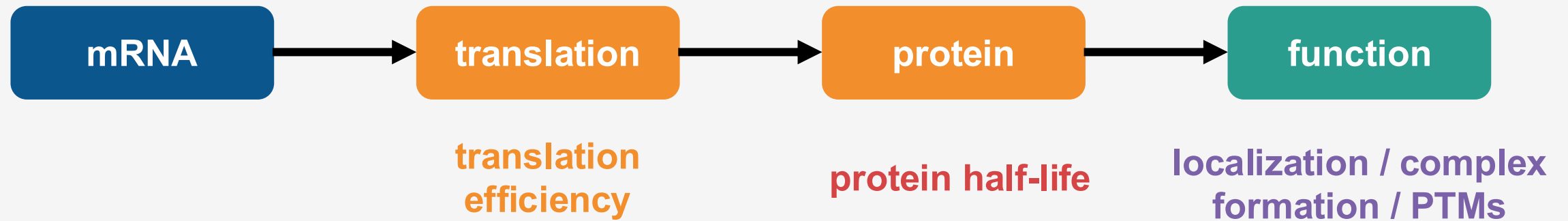
When RNA, protein and metabolites agree, and when disagreement is the most interesting result.

We will NOT re-teach PCA, volcano plots and FDR.

Instead: what is biologically different, what can go wrong, and how to connect the layers.

Proteomics

Why RNA and protein disagree (and why that is useful)



- A protein can change without its transcript changing.
- A transcript can change before the protein has time to respond.
- The same protein abundance can have very different activity states.

Discordance is not “bad data” by default - it can reveal regulation between omics layers.

Activity: RNA changed, what should the protein do?

Pairs, 2 minutes. Predict the protein-level outcome.

Gene A

RNA $\uparrow$ 8×

Protein half-life: 4 days

Sample collected 6 h after
perturbation

Protein?

Gene B

RNA unchanged

Protein rapidly degraded
after phosphorylation

Protein?

Gene C

RNA $\uparrow$ 2×

Translation strongly inhibited

Protein?

Activity: RNA changed, what should the protein do?

Pairs, 2 minutes. Predict the protein-level outcome.

Gene A

RNA $\uparrow$ 8×
Protein half-life: 4 days
Sample collected 6 h after
perturbation

Protein?

Gene B

RNA unchanged
Protein rapidly degraded
after phosphorylation

Protein?

Gene C

RNA $\uparrow$ 2×
Translation strongly inhibited

Protein?

mRNA

translation

protein

function

translation
efficiency

protein half-life

localization / complex
formation / PTMs

Transcriptomics measures regulatory response. Proteomics measures the molecular machinery that survived translation + turnover

Proteomics: One “protein” is not one thing

ABUNDANCE

How much protein is present?

Most common entry point for differential proteomics.

ISOFORMS

Different protein forms can have different functions or localization.

PTMs

Phosphorylation, acetylation, ubiquitination etc. can change activity without changing abundance.

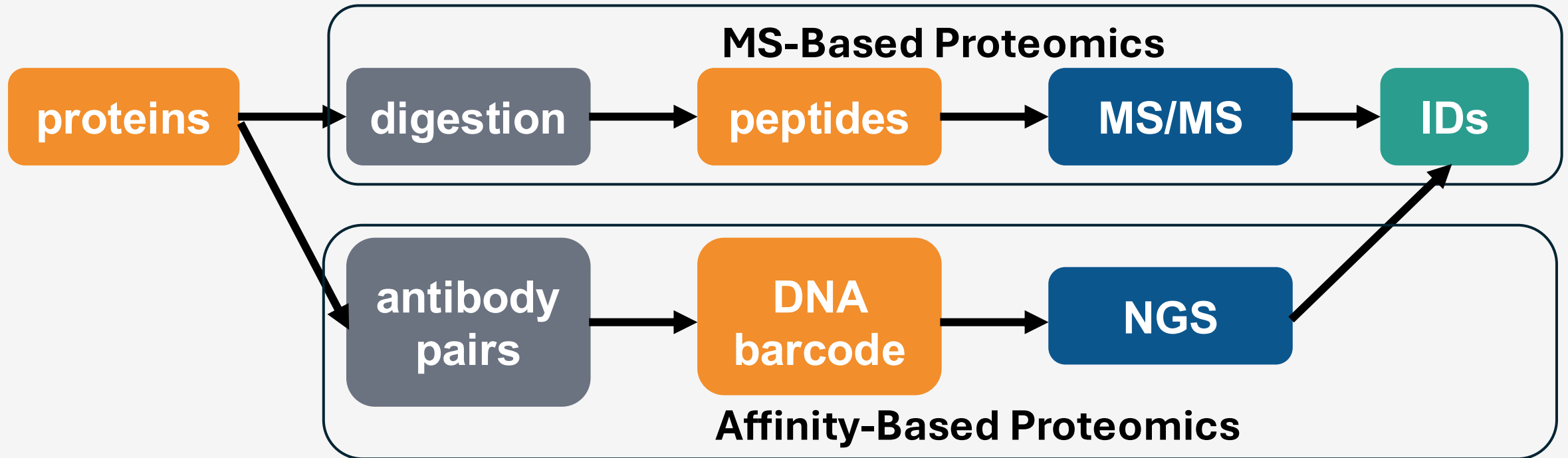
COMPLEXES

Function often depends on who the protein interacts with.

**Example: kinase abundance unchanged
phosphorylation ↑ strongly
pathway activity ↑**

Proteomics can move us closer to function than transcriptomics, but only if we know what “protein measurement” we actually have.

Measure or Infer Proteins?



Protein inference

Several proteins can share peptides. Which protein produced the peptide?

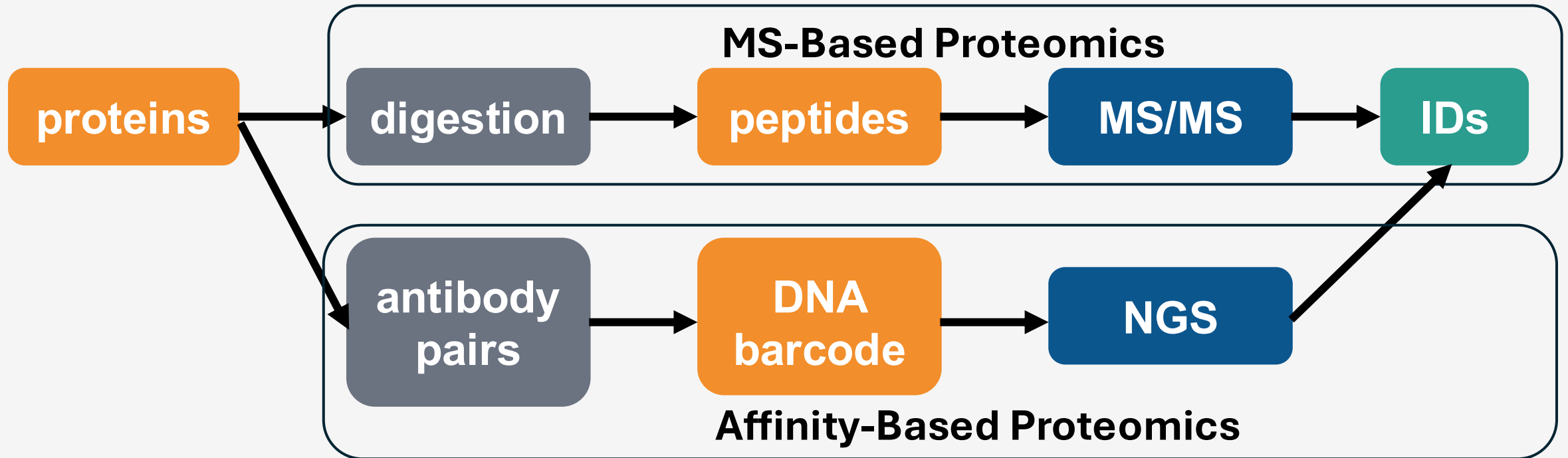
Missing values

A missing intensity can mean “not detected”, not necessarily “not present”.

Dynamic range

Highly abundant proteins can dominate; low-abundance signaling proteins are harder.

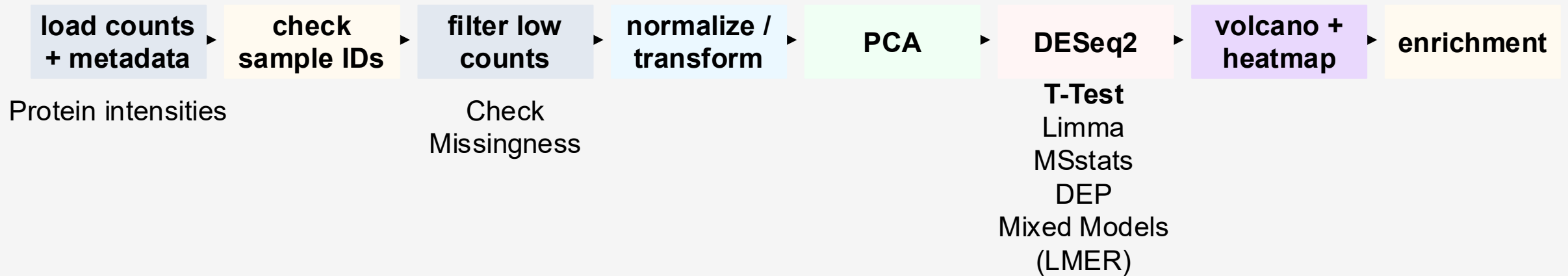
Measure or Infer Proteins?



Proteomics has an identification problem before it has a statistics problem.

Proteomics Pipeline

Proteomics pipeline is similar with Transcriptomics!



ENRICHMENT

GO / Reactome / KEGG

Similar to Transcriptomics

PROTEIN NETWORKS

STRING / interaction networks

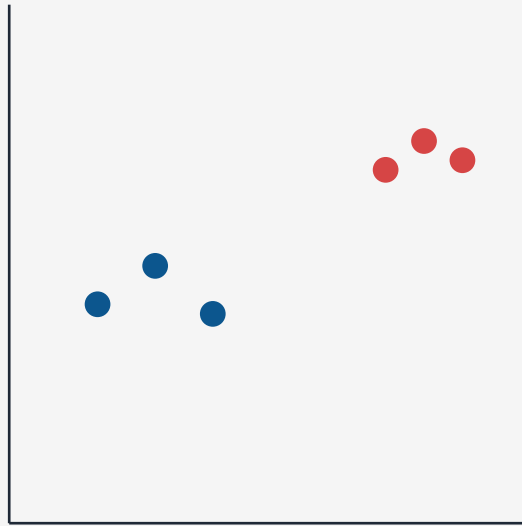
Do changed proteins form a connected module?

PTM ENRICHMENT

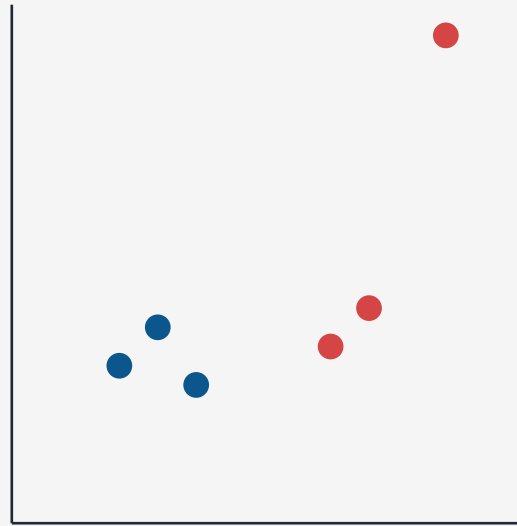
Kinase / phosphosite analysis
Which signaling programs may be active?

Proteomics QC: what would make you nervous?

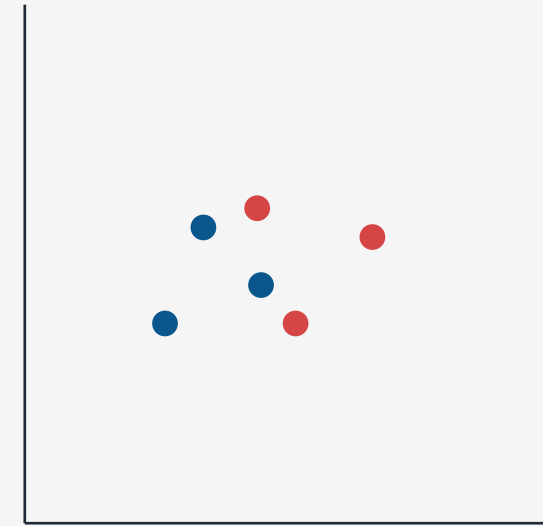
Quick vote: which plot would make you stop the analysis first?



A group separation



B one extreme sample



C no condition pattern

**In proteomics, also inspect missingness per sample/protein and intensity distributions
PCA alone is not enough.**

Online Tools

Analyst Suites Monash Proteomics



**Analyst Suites
MONASH PROTEOMICS**

A Compilation of Easy-To-Use Web Applications for
Proteomic Data Analysis and Visualisation

Powered by MPMP and MGBP

Expression Analysis



LFQ-Analyst

Requires MaxQuant (MQ) Output.

[Launch \(Latest\)](#) [Launch \(Published\)](#) [More](#)



Phospho-Analyst

Requires MaxQuant (MQ) Output.

[Launch](#) [More](#)



FragPipe-Analyst

Requires FragPipe Output.

[Launch \(Stable\)](#) [Launch \(Dev\)](#) [More](#)



TMT-Analyst

Requires Proteome Discoverer (PD) Output.

[Launch](#) [More](#)



DIA-Analyst

Requires Spectronaut or DIA-NN Output.

[Launch](#) [More](#)

Enrichment Analysis



Pathway-Analyst

Requires expression data (one from above analyst-suites apps or from Degust).

[Launch](#) [More](#)

ProteoRE

HeatMap (clustering and visualization)

DATA MANIPULATION AND VISUALIZATION

ID Converter (Human, Mouse, Rat)

Filter by keywords and/or numerical value

Venn diagram [J/Venn]

Volcano Plot create a volcano plot

GET FEATURES/ANNOTATIONS

Add protein features [UniProt, Human, Mouse, Rat]

Add protein features [neXtProt, Human]

Get expression profiles by (normal or tumor) tissue/cell type [Human Protein Atlas]

Add expression data [RNAseq or Immuno-assays] [Human Protein Atlas]

Build tissue-specific expression dataset [Human Protein Atlas]

Get MS/MS observations in tissue/fluid [Human Peptide Atlas]

Get unique peptide SRM-MRM method [Human SRM Atlas]

FUNCTIONAL ANALYSIS

Compute functional profiles (Human, Mouse) [goProfiles]

Classification and enrichment analysis (Human, Mouse, Rat) [clusterProfiler]

Biological theme comparison (Human, Mouse, Rat) [clusterProfiler]

Enrichment analysis (Human, Mouse, Rat) [topGO]

PATHWAY ANALYSIS

Pathway enrichment analysis [Reactome]

KEGG pathways mapping and visualization [PathView]

KEGG pathways mapping and coverage [PathView]

Build Protein Interaction network

ProteoRE



Welcome to ProteoRE !



ProteoRE.org will be unavailable for maintenance purpose from Monday, September 21st to Thursday, September 24th 2026.

More details on Migale web page Thank you for your understanding.



What is ProteoRE ?

ProteoRE (Proteomics Research Environment) is a Galaxy-based platform for the functional analysis and the interpretation of proteomics and transcriptomics data in biomedical research; ProteoRE allows researchers to apply a large range of bioinformatics tools and tailored workflows on their own data using Galaxy capabilities. ProteoRE comprises **21 tools** organized into four subsections: i) data manipulation and visualization; ii) get features/annotation; iii) functional analysis; and iv) pathway analysis along with graphical representation. This version also includes Galaxy generic tools and some tools from the GalaxyP project that should be of great help for (re)processing MS-based proteomics data (e.g. peptide/protein identification).

ProteoRE new release (24th June 2024) - release notes

- ProteoRE server: CPU and memory resources have been extended and jobs management configuration revisited to better satisfy user needs
- Update of resources used by ProteoRE tools (more details here):
 - ID mapping files have been updated ("ID-converter")
 - Expression data sources from Human Protein Atlas ("Get expression profiles", "Add expression data", "Build tissue-specific expression dataset")
 - PPI data from BioGRID ("Build Protein Interaction network")
- ProteoRE tutorials and workflows updated:
 - Tutorial 1: Annotating a protein list identified by LC-MS/MS experiments (release 2.0) (from PMID: 29189203)
 - Tutorial 2: Biomarkers candidate identification (from PMID: 30852829)
 - Workflow for the Selection of Pancreatic Cancer Biomarkers (from PMID: 31538697)
 - Annotation and GO Enrichment Analysis of a List of Differentially Expressed Proteins (from PMID: 34236662)

How to access ProteoRE ?

This public site is meant for academic usage; for commercial usage, contact us). ProteoRE can currently be used as:

- Unregistered user:** you can analyze your data (e.g. proteomics identification results in tab format) using the available tools. Allocated storage space is limited to 20 Mb and data/history/analyses are not conserved between sessions.
- Registered user** (password-protected access): we recommend to create a user account. Up to 20 Gb storage space is offered to registered users, along with access to shared resources (data, histories, workflows...). To create an account: click on "Account registration or login" on the main menu bar (upper part of the ProteoRE central panel), click on "registration" and fill out the form. Once the form is submitted, your account is created.

ProteoRE Tutorials available in Galaxy Training Network:

- Annotating a protein list identified by LC-MS/MS experiment
- Strategy to select biomarker candidates

Metabolomics

Metabolomics: a biochemical snapshot

FAST

Metabolites can change in seconds or minutes.
They can respond much faster than transcription.

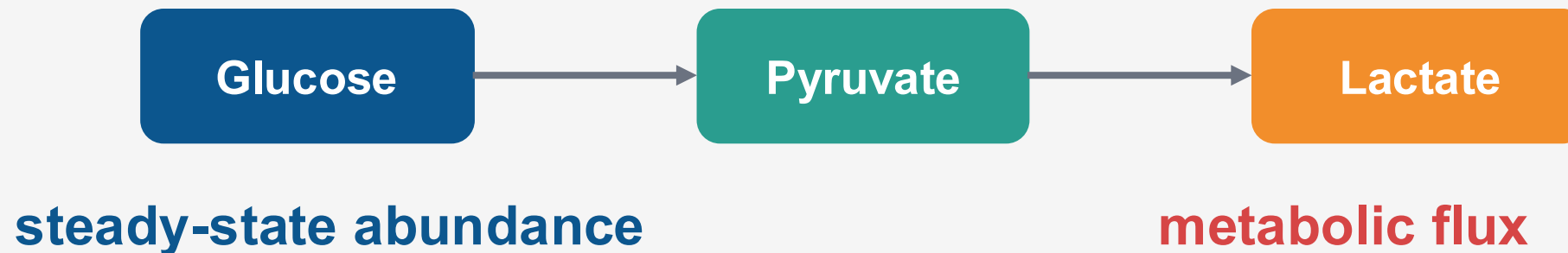
CONTEXTUAL

Diet, drugs, microbiome, fasting, circadian rhythm, handling and storage can all matter.

CLOSE TO PHENOTYPE

Metabolites are substrates/products of biochemical reactions and often reflect pathway state.

Think about Abundance vs Flux!



Concentration question

"Is lactate higher?"

Flux question

"Is more glucose carbon being converted into lactate?"

Targeted vs Untargeted

TARGETED

Measure a defined metabolite panel.

- + better identification confidence
- + often more quantitative
- + easier biological interpretation
- limited to what you decided to measure

UNTARGETED

Measure thousands of spectral features.

- + discovery potential
- + broad chemical space
- many features remain unknown
- one metabolite may create several peaks

**Untargeted metabolomics often begins with “features”
metabolite identity is a separate analytical step.**

Questions for Metabolomics

EXPOSURE

Drug metabolites, diet-derived compounds, environmental chemicals.

ENERGY / REDOX

ATP/ADP, NAD-related metabolites, TCA intermediates, glutathione.

PATHWAY OUTPUT

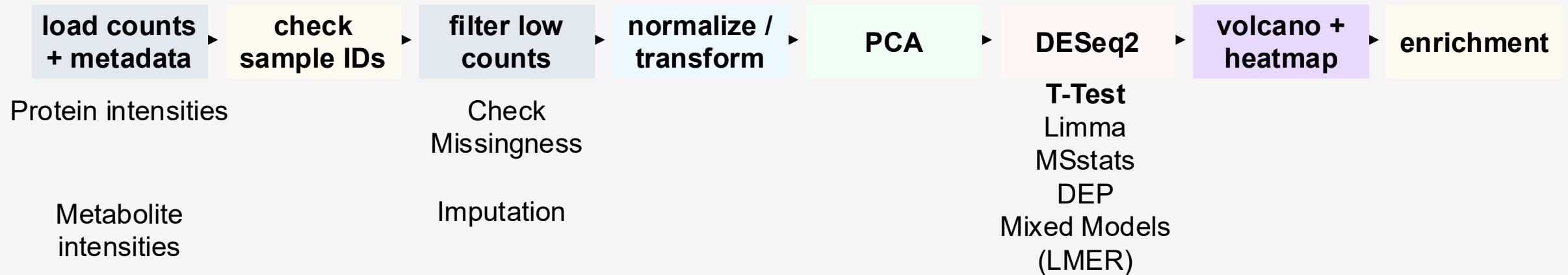
Accumulation or depletion of substrates/products can reveal pathway bottlenecks.

A transcript can tell you an enzyme may be induced. A metabolite can tell you the biochemical system actually changed.

But remember: concentration $\neq$ flux. For flux, think isotope tracing.

Metabolomics Pipeline

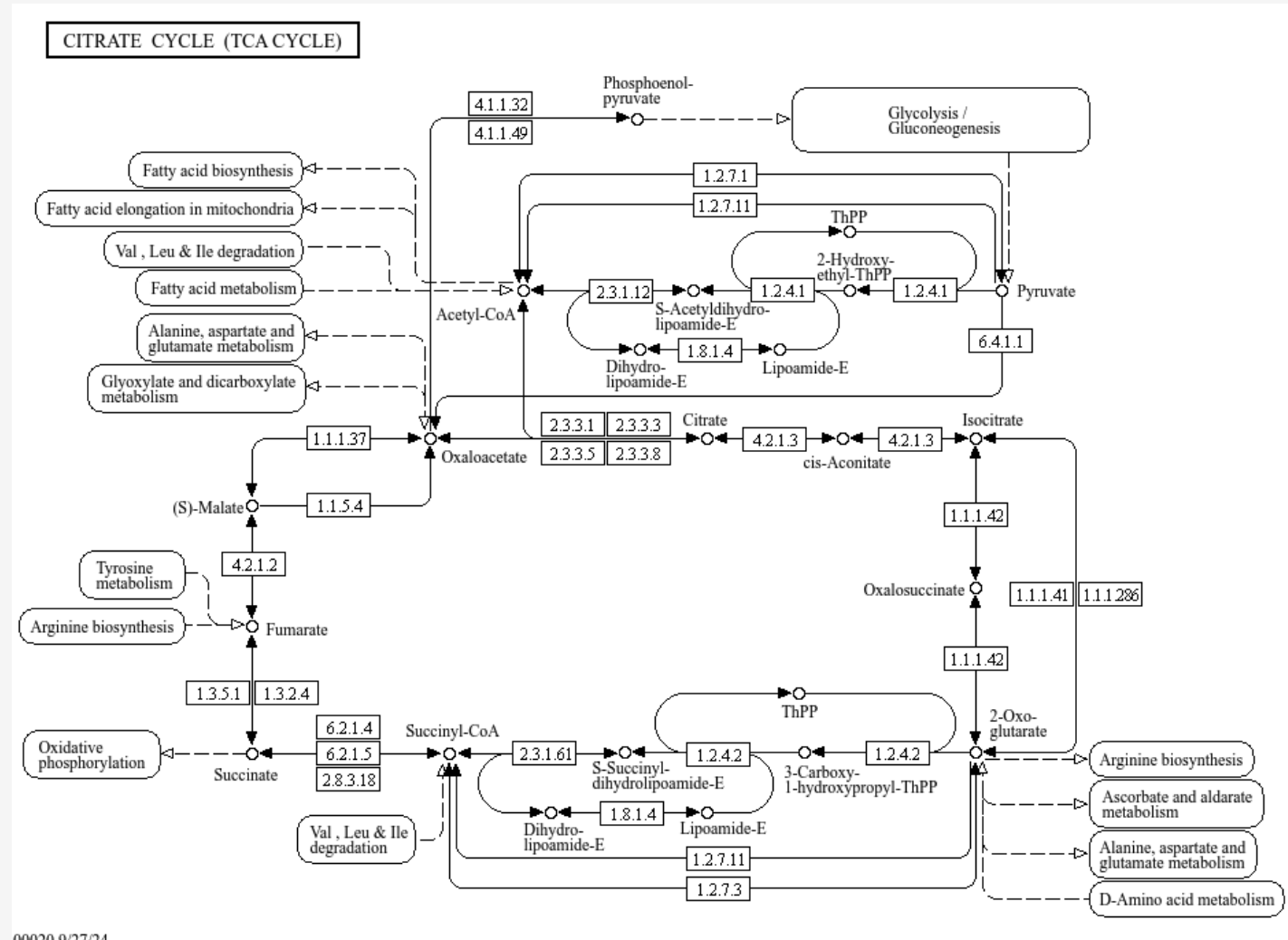
Metabolomics pipeline is similar with Proteomics! Both starts from continuous abundance



ENRICHMENT

GO / Reactome / **KEGG/**
MetaboAnalyst

KEGG for Metabolomics



00020 9/27/24

MetaboAnalyst

MetaboAnalyst 6.0 - from raw spectra to biomarkers, patterns, functions and systems biology

Module Overview

Input Data Type	Available Modules (click on a module to proceed, or scroll down to explore a total of 18 modules including utilities)				
LC-MS Spectra (mzML, mzXML or mzData)			Spectra Processing [LC-MS w/wo MS2]		
MS Peaks (peak list or intensity table)		Peak Annotation [MS2-DDA/DIA]	Functional Analysis [LC-MS]	Functional Meta-analysis [LC-MS]	
Generic Format (.csv or .txt table files)	Statistical Analysis [one factor]	Statistical Analysis [metadata table]	Biomarker Analysis	Dose Response Analysis	Statistical Meta-analysis
Annotated Features (metabolite list or table)		Enrichment Analysis	Pathway Analysis	Network Analysis	
Link to Genomics & Phenotypes (metabolite list)			Causal Analysis [Mendelian randomization]		

Things can get exciting: Omics Disagreement



The most informative multi-omics result is often not agreement. It is a biologically interpretable mismatch.

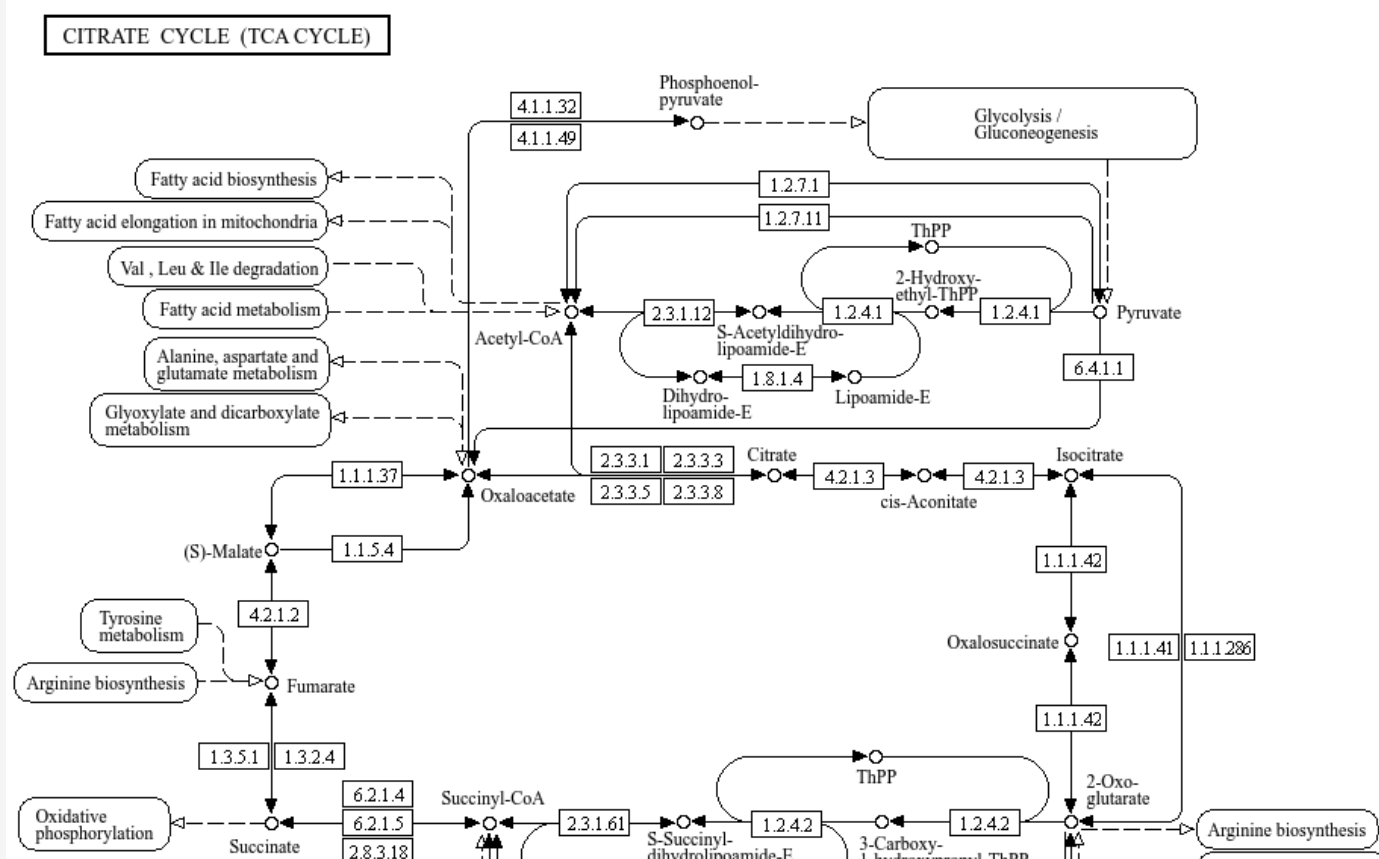
MultiOmics Integration

Simple ways to integrate Omics

1. PATHWAY CONVERGENCE

Analyze each omics separately.
Ask whether changed genes/proteins/metabolites point to the same pathway.

Best for beginners.



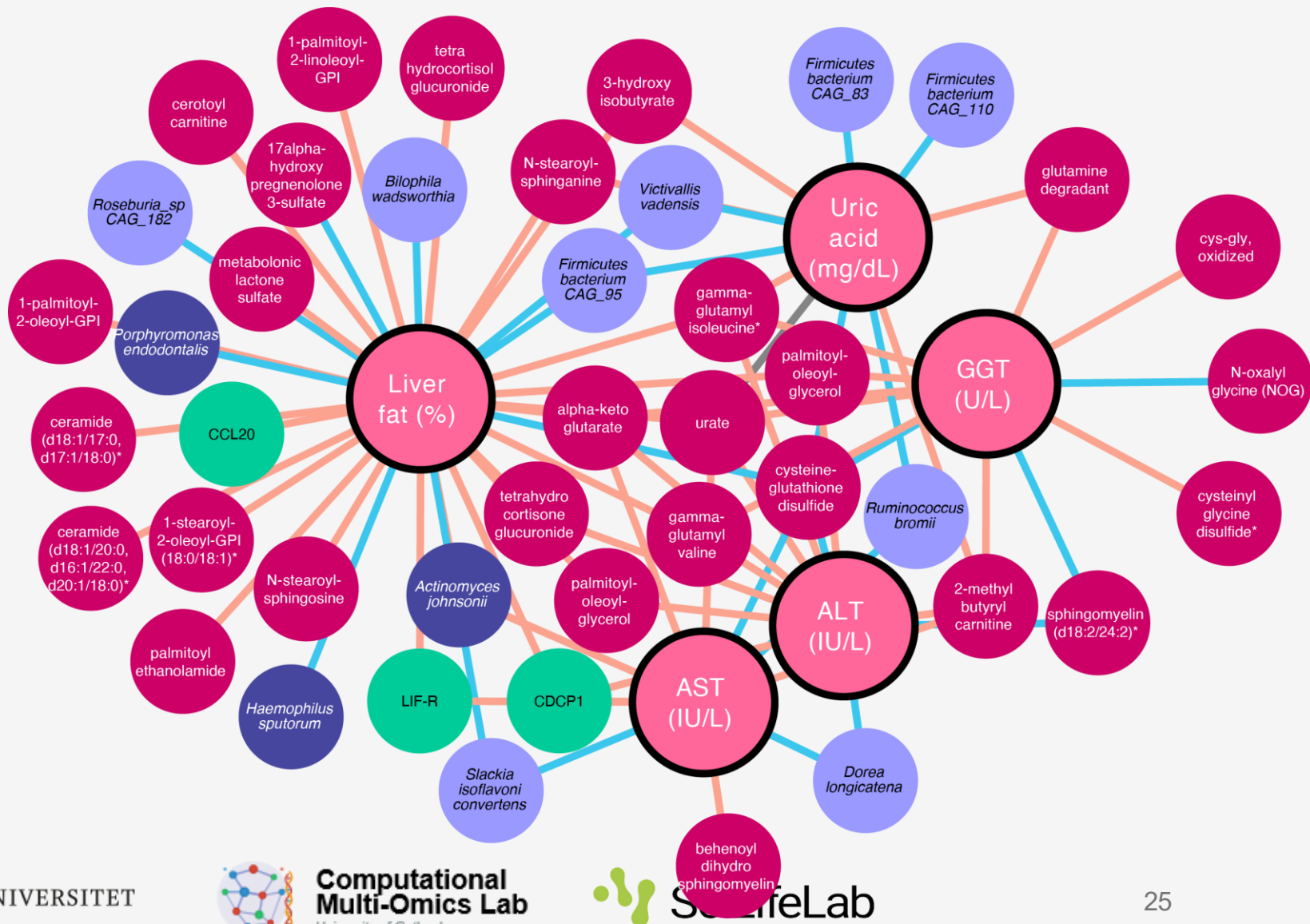
Star simple. Integration should answer a biological question, not just combine matrices because you have them.

Simple ways to integrate Omics

2. FEATURE CORRELATION

Match samples across layers.
Correlate gene $\leftrightarrow$ protein, enzyme
 $\leftrightarrow$ metabolite, protein $\leftrightarrow$ metabolite.

Good for hypothesis generation.



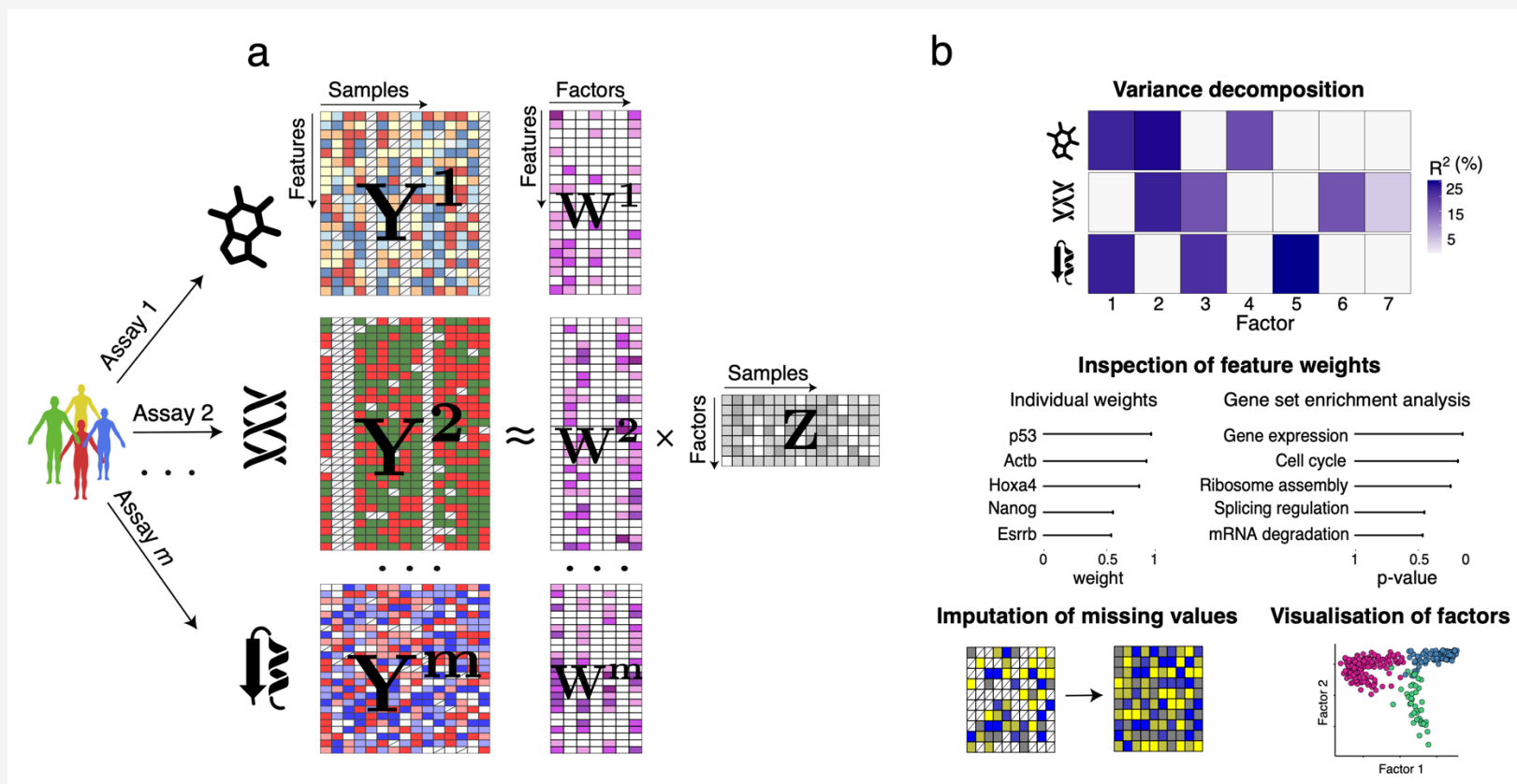
Simple ways to integrate Omics

3. JOINT MODELS

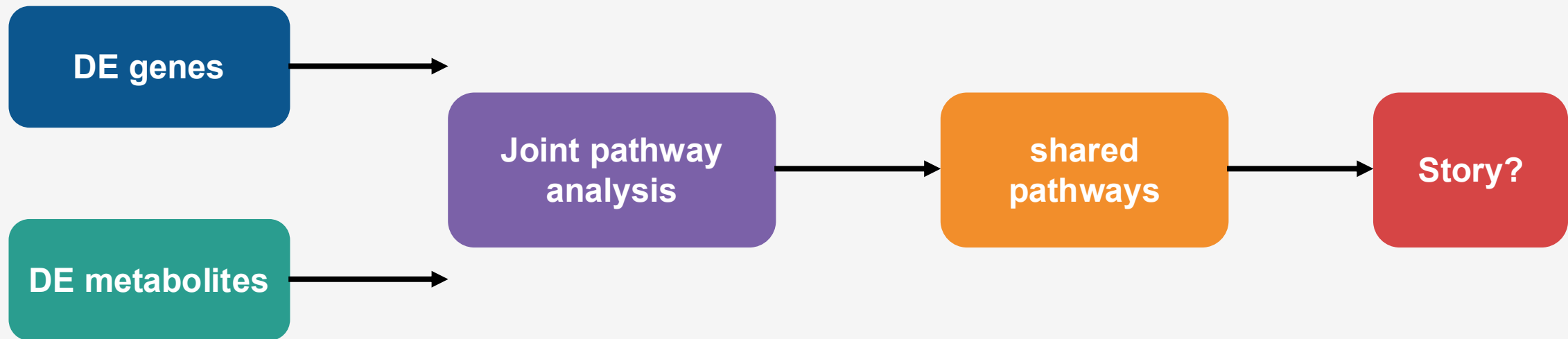
Integrate matrices together: latent factors, sparse models, networks, machine learning.

Powerful, but harder to interpret.

MOFA or MixOmics



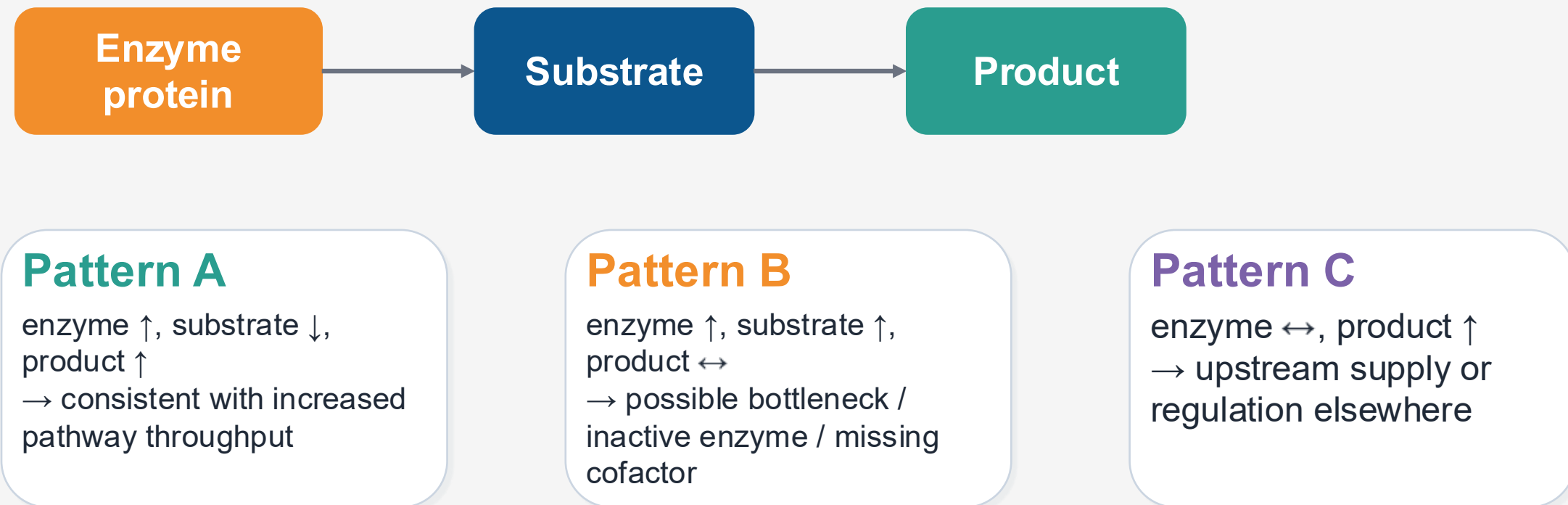
Transcriptomics + Metabolomics



- Example: collagen/fibrosis genes $\uparrow$ + TCA/redox metabolites altered $\rightarrow$ stronger evidence for metabolic remodeling.
- But a “shared pathway” is not proof that the gene change caused the metabolite change.
- Use time, perturbation or targeted validation to move toward mechanism.

Integration strengthens interpretation when independent molecular layers converge on the same biology

Proteomics + Metabolomics



Multi-omics becomes mechanistic when you interpret molecules in reaction context, not only by correlation.

PaintOmics 4

PaintOmics AI

Job view Storage Tools More

Load example Run PaintOmics Reset Sign in

PaintOmics AI

Paint every omic layer onto the pathway that explains it — across KEGG, Reactome and MapMan, for species from every biological kingdom.

AI-powered pathway interpretation

The **PaintOmics AI agent** writes an interpretation of your ranked pathways — a draft, for you to check.

- queries your values
- runs its own literature searches
- verifies every citation

Load an example Documentation GitHub Contact

AI interpretation

OmicsAnalyst

OmicsAnalyst 2.0

A unified platform for statistical, functional, and causal analysis of multi-omics data.

Get Started →

This platform is freely available to everyone, including commercial users.

So many tools and methods! How to choose?

Do Pathways Agree?

1. PATHWAY CONVERGENCE

Analyze each omics separately.
Ask whether changed genes/proteins/metabolites point to the same pathway.

Best for beginners.

Which features co-vary?

2. FEATURE CORRELATION

Match samples across layers.
Correlate gene $\leftrightarrow$ protein, enzyme $\leftrightarrow$ metabolite, protein $\leftrightarrow$ metabolite.

Good for hypothesis generation.

I need to predict phenotypes

3. JOINT MODELS

Integrate matrices together: latent factors, sparse models, networks, machine learning.

Powerful, but harder to interpret.

Take-home messages

PROTEOMICS

Closer to molecular machinery.
Think peptides, protein inference, missingness, PTMs and networks.

METABOLOMICS

Closer to biochemical phenotype.
Think annotation, scaling, exposure, pathway output and flux.

MULTI-OMICS

Agreement is useful.
Disagreement can be even more informative if biology explains it.

The statistical toolkit is transferable. The interpretation is omics-specific.

Ask: What does this layer measure that the previous layer could not?

[illegible]

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